SCS 3204: AUTOMATA THEORY

Lecture V:
Finite Automata

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Recap

- A precise definition aids in carrying out proof
- A concise definition assists understanding
- Recursive definitions are precise and concise
- Regular expressions : another method of defining languages.
- Finite Automata: yet another method for defining languages.

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Terminologies

- Finite Automata
- Finite State Automata
- Finite State Machines
- Deterministic/Non-deterministic Finite Automata
- Note:

  Automaton = Singular

  Automata = Plural

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Finite State Automata (FA)

- Key ideas:

  State

  Finite

  Network

  Automaton

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Definition - State

- Substances, people and physical machines are often said to be in one state or another.
- H_2O : frozen state (ice) ⇒ liquid state (water) ⇒ gaseous state (water vapour)

  under the right conditions (subjected to certain inputs), H_2O will change from one state to another.
- People? Happy ⇒ depressed ⇒ excited ⇒ bored ⇒ ... ⇒ dead state

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Definition - State

- Machines? e.g. Light switch has two states

  ON (corresponds to switch lever being up) and

  OFF (corresponds to switch lever being down)
- When switch lever is up (ON state), we can push it down and when its down (OFF state), we can push it up.
- The possible TRANSITIONS from one state to another are modelled using arcs from one state to another.
- Each arc is labeled with the INPUT that causes that particular change of state.

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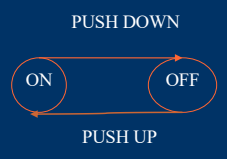
States and Transitions of an On-Off Light Switch



Note: Illegal inputs and transitions are left out of the model e.g. Pushing down when the switch is off is illegal as it would break the machine.

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Example – Old Car Fan

- Fan has exactly 4 settings (STATES) i.e. OFF, LOW, MED and HIGH.
- One changes the state by turning the knob Left (anticlockwise) or Right (clockwise).
- Fan does not allow direct transition from OFF to MED or HIGH, from LOW to HIGH, from MED to OFF, or from HIGH to LOW or OFF.

Draw a diagram of the state model of this old car fan.

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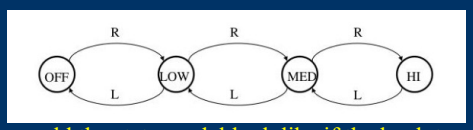

How would the state model look like if the knob turns only in the clockwise direction?

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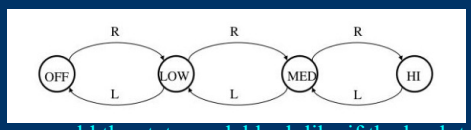

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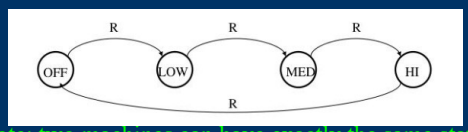

Note: two machines can have exactly the same states, but different inputs and transitions!

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Definitions

- FINITE => Number of states and inputs are both finite!
- NETWORK => Graph-like structures of nodes linked together with directed, labeled arc-transitions.
- AUTOMATON => change of state is totally controlled by the input i.e. Determination of what state is next is automatic (mechanical)

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Definition

- A finite automaton (FA) consists of:
  - A finite set of states one of which is the **start state** and some (maybe none) are **final states**
  - An alphabet $\sum$ of possible **input** letters that form input strings that are read one letter at a time
  - A finite set of **transitions** that tell for each state and for each letter of the input alphabet, which state to go to next.

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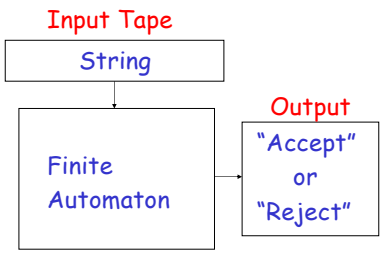
Deterministic Finite Automaton (DFA)



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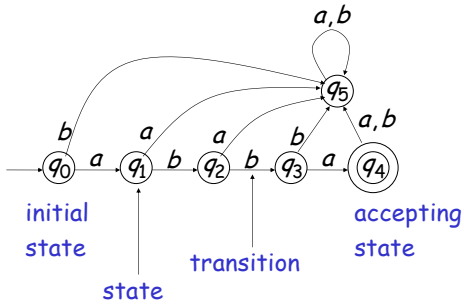
Transition Graph



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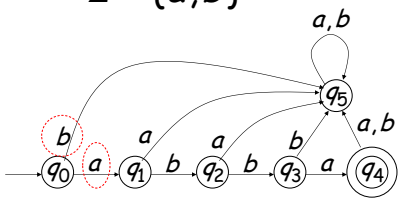
Alphabet $\Sigma = \{a,b\}$



For every state, there is a transition for every symbol in the alphabet

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How it works

- Presented with an input string at start state
- Reads input string letter by letter beginning with the leftmost letter
- Each letter read causes the automaton to transit to a new state (where the next letter will be read)
- The sequence ends when the last input letter has been read
- If the end occurs at a final state then the input string has been accepted by the automaton

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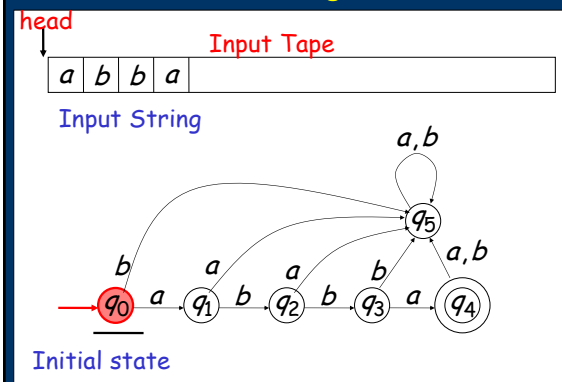
Initial Configuration



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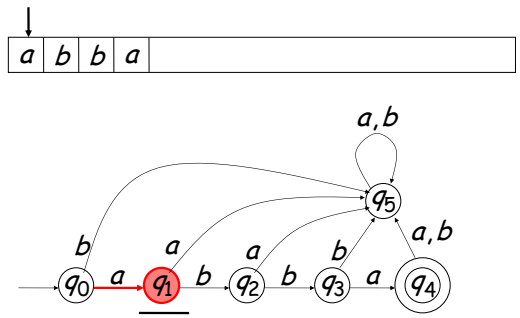
Scanning the Input



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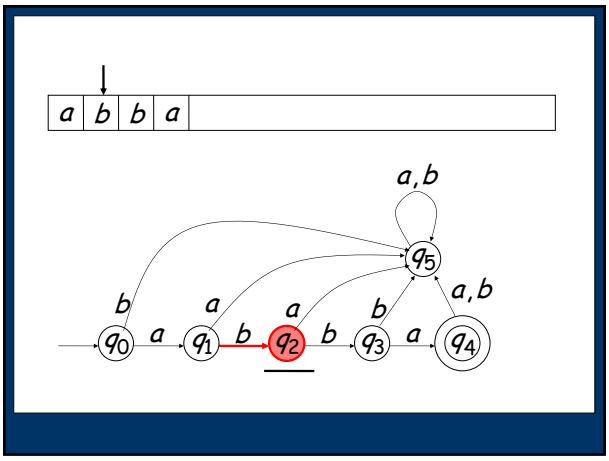

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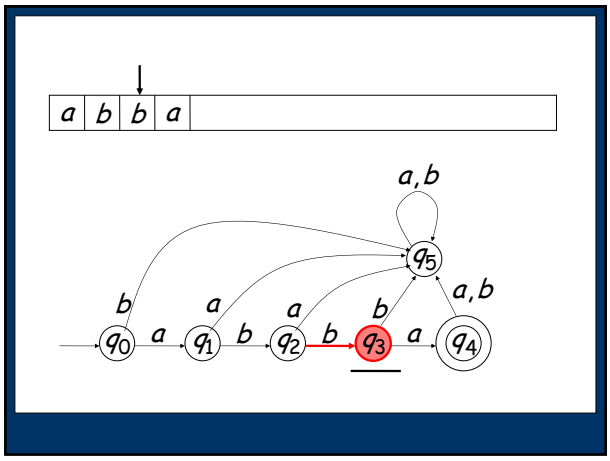

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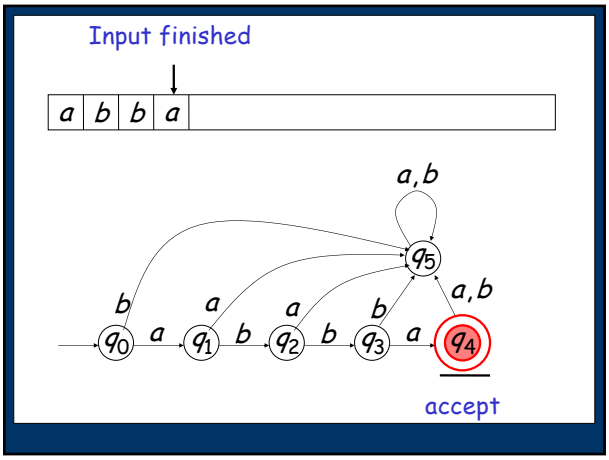

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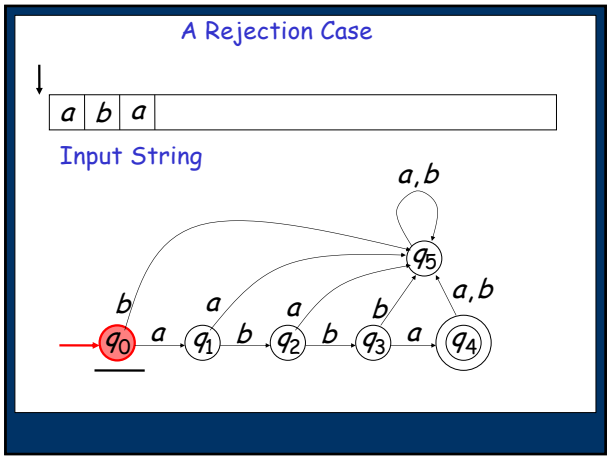

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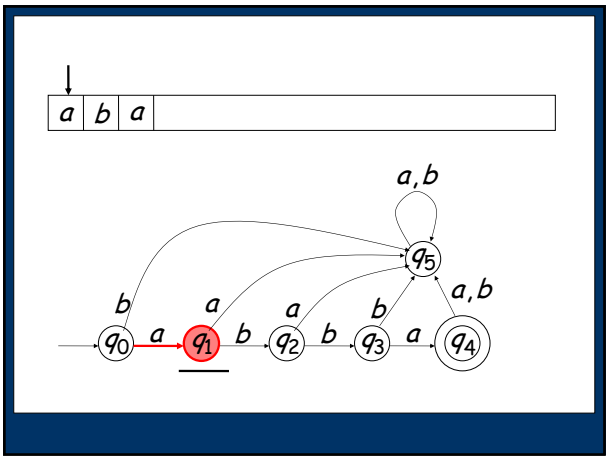

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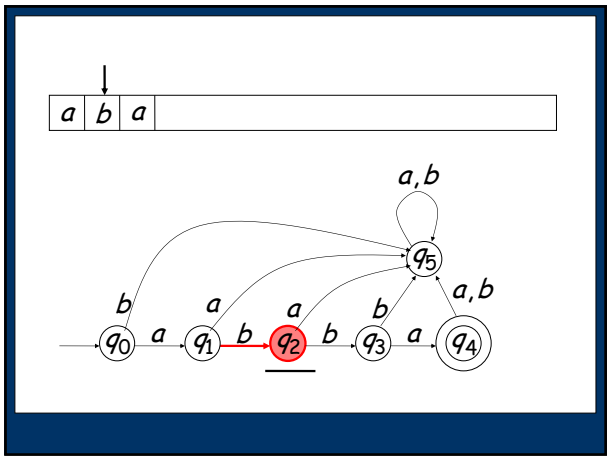

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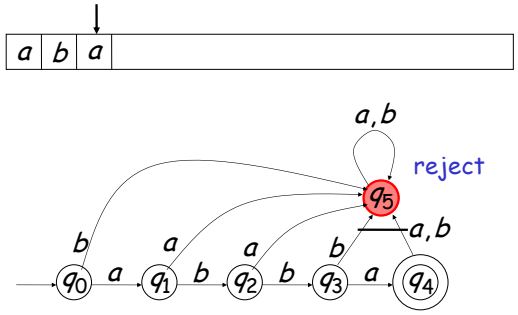
Input finished



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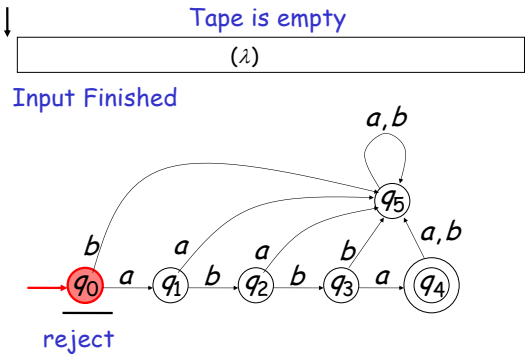
Another Rejection Case

Tape is empty



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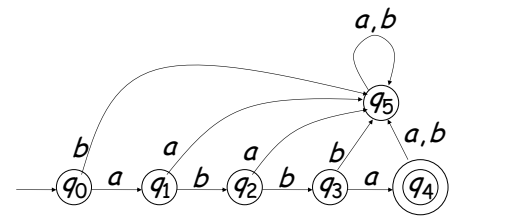
Language Accepted: $L = \{abba\}$



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To accept a string:

all the input string is scanned
and the last state is accepting

To reject a string:

all the input string is scanned
and the last state is non-accepting

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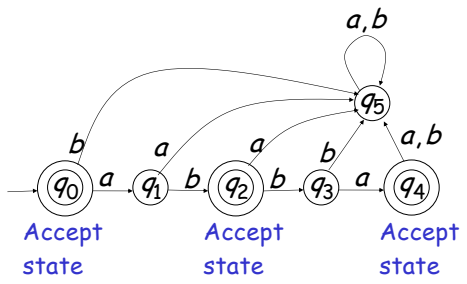
Another Example

$L = \{\lambda, ab, abba\}$



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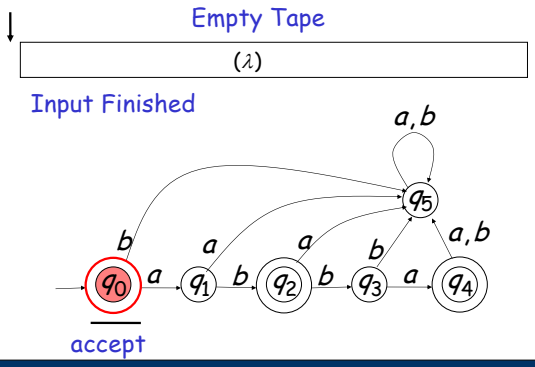
Empty Tape



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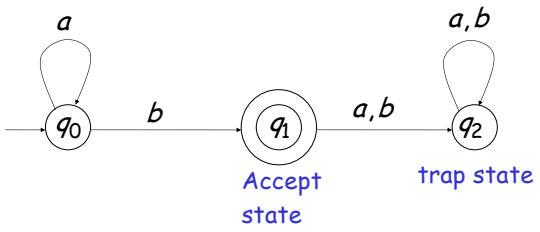
Another Example



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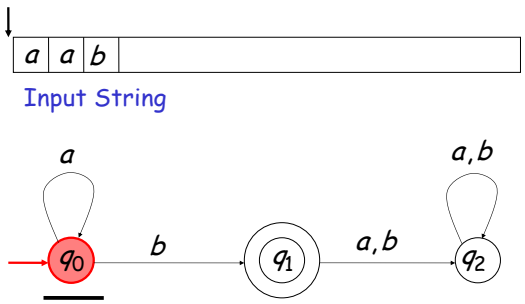

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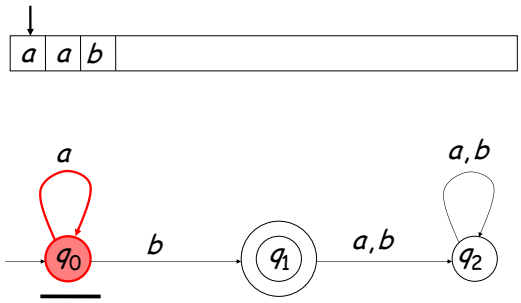

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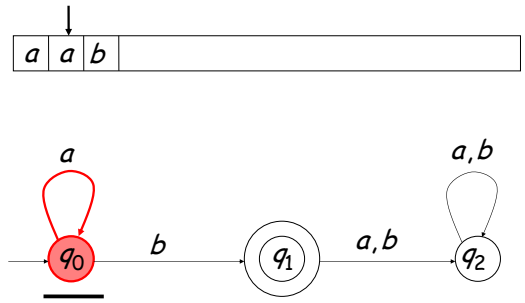

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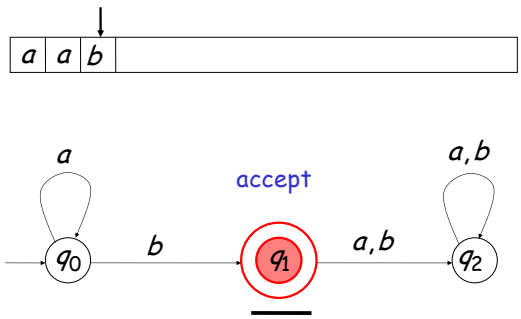

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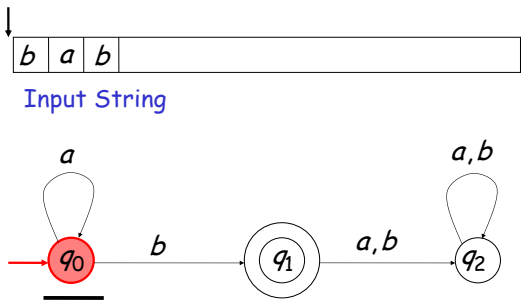

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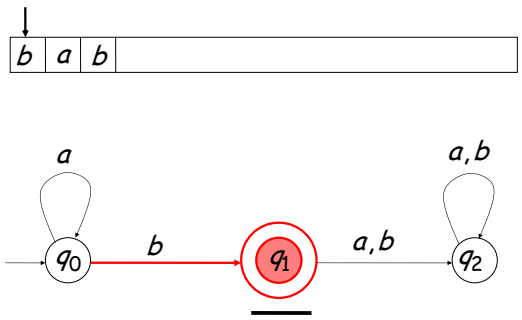

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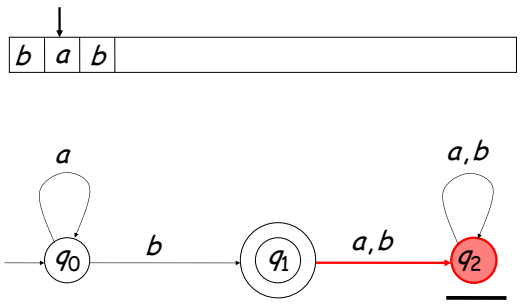

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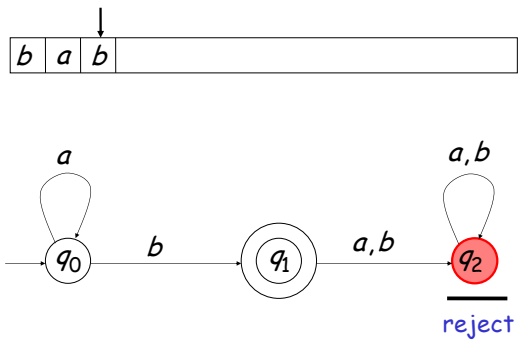
Input finished



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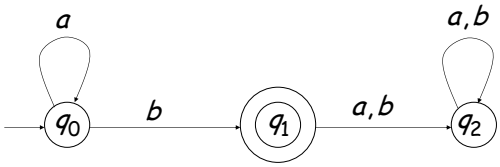
Language Accepted: $L = \{a^n b : n \geq 0\}$



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Connection with Set Theory

- The set of all strings that terminate at a final state is called the language defined by the finite automaton
- Any string that does not end at a final state is not accepted, and does not belong to the language

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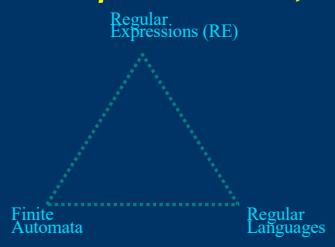
The relationship between FA, RE and RL



- Any RE that does not use the memory feature can be implemented as a FA; likewise, any FA can be described with a RE.
- Both FA and RE can be used to describe regular languages.

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Representation of FAs

- Finite automata can be represented using tables or transition diagrams:
  - The table representation makes it easier to perform proofs
  - The transition diagram makes it easier to process strings

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Formal Definition

Deterministic Finite Automaton (DFA)

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q : set of states

Σ : input alphabet $\lambda \notin \Sigma$

δ : transition function

q_0 : initial state

F : set of accepting (final, terminal) states

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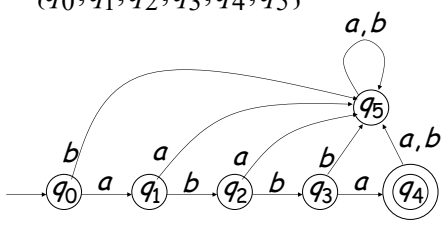
Set of States

Example

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$$



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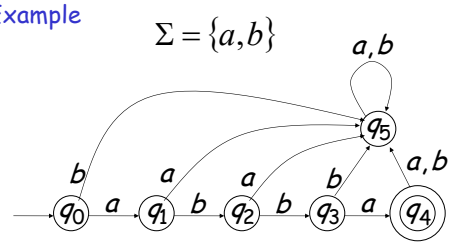
Input Alphabet

Σ

$\lambda \notin \Sigma$:the input alphabet never contains λ

Example

$$\Sigma = \{a, b\}$$



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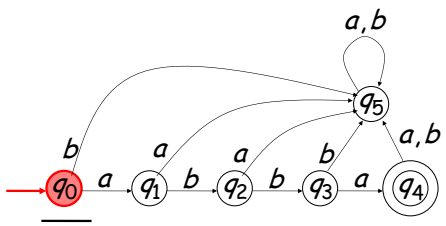
Initial State

q_0

Example



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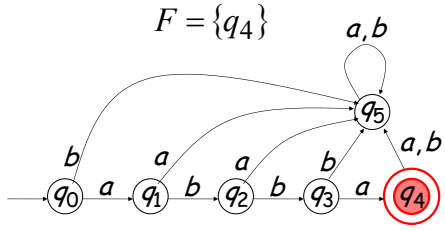
Set of Accepting States

$F \subseteq Q$

Example

$$F = \{q_4\}$$



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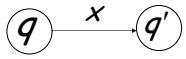
Transition Function

$$\delta : Q \times \Sigma \rightarrow Q$$

$$\delta(q, x) = q'$$



Describes the result of a transition from state q with symbol x

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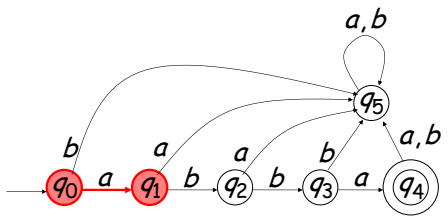
Example:

$$\delta(q_0, a) = q_1$$



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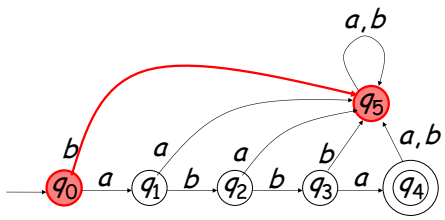
$$\delta(q_0, b) = q_5$$



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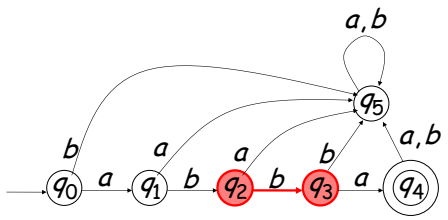
$$\delta(q_2, b) = q_3$$



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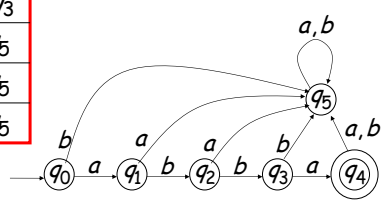
Transition Table for δ

symbols

states

δ	a	b
q_0	q_1	q_5
q_1	q_5	q_2
q_2	q_5	q_3
q_3	q_4	q_5
q_4	q_5	q_5
q_5	q_5	q_5



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Another Example: Table Representation

- Remember the regular expression:
  $(a + b)^* b\ (a + b)^*$
- What language does it represent?
- We can represent the same information in a tabular finite automaton

Input letter → State ↓	a	b
Start x	y	z
y	x	z
Final z	z	z

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Example

- Using the state transition table, is the string aaba in the language?

Note: for earlier transition table:

- The letter a keepscircuiting in the table without making any progress
- The letter b always puts the machine in a final state
- Once in the final state, the machine does not change state but remains there, regardless of received input

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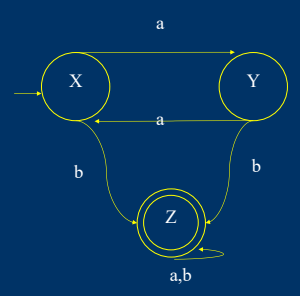
Transition Diagram (Network)



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Language Accepted by DFA

Language of DFA : M

it is denoted as $L(M)$ and contains all the strings accepted by M

We say that a language L' is accepted (or recognized) by DFA M if $L(M) = L'$

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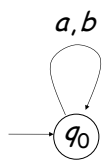
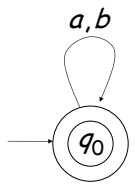
More DFA Examples

$\Sigma = \{a,b\}$



$L(M) = \{\ \}$

Empty language



$L(M) = \Sigma^*$

All strings

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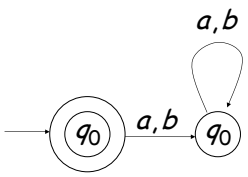
$\Sigma = \{a,b\}$



$L(M) = \{\lambda\}$

Language of the empty string

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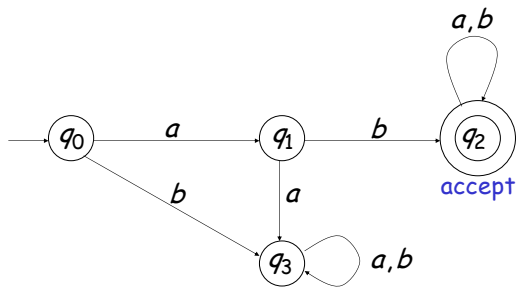
$\Sigma = \{a,b\}$

$L(M)$



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Example

- Given "sheep language" which is defined as any string from the following (infinite) set:

  baa
  baaa
  baaaa
  baaaaa
  baaaaaa.....
- What is the regular expression defining sheeptalk?
- Develop a tabular finite automaton for sheeptalk.
- What is the corresponding FA for sheeptalk?

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Example

- Design an FA which accepts L (defined over the alphabet $\sum = \{a\ b\}$) which is described by the following regular expressions:

  $(a + b)(a + b)^*$
  $(a + b)^*$
  $b\ (a + b)^*\ b$
  $b\ (ab + b)\ b$

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Example

Given Language $L = \{a^m b^n : m,n > 0\}$ which is defined over the alphabet $\sum = \{a\ b\}$

- What is the descriptive definition of L?
- What is the regular expression defining L?
- Develop a tabular finite automaton for L.
- What is the corresponding FA for L?

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Example

- Design a FA which accepts the language L which is defined over the alphabet $\sum = \{a\ b\}$ and whose words do not contain three consecutive b's.

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Example

- Obtain the FA that accepts/recognizes the language L which is defined over the alphabet $\sum = \{a\ b\ c\}$, where $L = (a + b + c)^*$ and each word contains the pattern *abac*.

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Exercises

Develop finite automata for the following languages which are defined over the alphabet : $\sum = \{a\ b\}$

1. aaaaa
2. $b(a + b)^*$
3. $a(a + b)^*b + b(a + b)^*a$
4. $L = \{b^m a b^n : m,n > 0\}$
5. L defined over {a,b} containing strings starting with prefix ab

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